

AI-System for Short Circuit and Lineman Location Alerts

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Abstract—In electrical power distribution systems, short-circuit faults are a significant contributor to equipment damage, power outages, and safety hazards. Conventional manual inspection takes a lot of time, puts linemen in danger, and is ineffective at finding problems. Current IoT-based solutions enhance detection, but they lack real-time visual confirmation, require costly hardware, and require complicated installation. This paper presents an AI-Based Short Circuit Detection and Lineman Alert System that utilizes the YOLOv11 deep learning model and a standard webcam to detect visual fault indicators such as sparks, smoke, and flashes in real time. A custom dataset consisting of real and simulated short-circuit images was created and used to train the model for accurate detection. The system employs OpenCV for real-time video processing and integrates with a Flask-based web dashboard to generate structured alerts containing the detected fault image, GPS-based location, and timestamp. The dashboard also supports alert management and event logging for maintenance analysis. The proposed sensor-free, vision-based approach reduces deployment cost, improves lineman safety, enables rapid fault localization, and provides an efficient and scalable solution for intelligent power distribution monitoring.

Keywords—Short Circuit Detection, YOLOv11, Computer Vision, Deep Learning, OpenCV, Flask, Power Distribution, Real-Time Monitoring, Lineman Safety, Artificial Intelligence.

I. INTRODUCTION

One of the most important problems with electrical power distribution systems is short-circuit faults, which frequently result in equipment damage, power outages, and major safety risks. Rapid urbanization and expanding distribution networks have increased the complexity of fault monitoring, making timely detection and accurate localization essential for ensuring uninterrupted power supply and operational safety. Traditional fault detection methods rely heavily on manual inspection by linemen, which is slow, labor-intensive, and exposes personnel to dangerous working conditions, especially during adverse weather and high-voltage environments.

A number of IoT-based monitoring systems that use sensors to identify anomalous electrical parameters have been introduced to address these issues. While effective to an extent, such

systems require costly hardware, complex installations, continuous network connectivity, and frequent maintenance. Furthermore, the majority of current solutions do not provide real-time visual confirmation of errors, which restricts situational awareness and could postpone remedial action. This paper proposes an AI-Based Short Circuit Detection and Lineman Alert System that employs a standard webcam and the YOLOv11 deep-learning model to detect visual indicators of short circuits, including smoke, sparks, and flashes, in real time. The system captures fault images, determines location information, and generates structured alerts for linemen through a web-based interface. The proposed sensor-free and vision-based approach reduces deployment cost, improves safety, and enables faster response, offering a scalable and practical solution for modern power distribution monitoring.

In addition, the system supports automatic event logging, where each detected short-circuit incident is recorded along with the captured image, timestamp, and location details. Electrical authorities can use this historical data for post-event analysis, fault trend identification, and preventive maintenance planning. By combining real-time visual monitoring with intelligent alerting and data storage, the proposed system enhances decision-making and reduces outage duration. The system can be deployed in both urban and rural power distribution networks due to its lightweight architecture, low hardware requirements, and flexibility for future improvements.

Furthermore, the proposed approach improves lineman safety by minimizing the need for physical patrolling of distribution lines. Early and precise fault identification reduces exposure to hazardous environments, thereby lowering the risk of accidents and operational delays. The availability of real-time visual evidence makes it possible for maintenance staff to remotely determine the severity of a fault and prepare the necessary corrective actions before they arrive at the site, leading to more effective restoration processes.

Moreover, the vision-based architecture provides flexibility and scalability that traditional sensor-based systems often lack. The system can be easily integrated with existing infrastructure using commonly available hardware such as webcams and standard computing devices. Its modular software design allows seamless incorporation of additional features, including advanced analytics, model retraining, and integration with smart grid platforms. This adaptability makes the proposed solution a promising step toward intelligent and automated power distribution fault management.

With the increasing demand for reliable and uninterrupted power supply, distribution networks require intelligent monitoring solutions that can respond instantly to faults. Conventional approaches lack adaptability to dynamic operating conditions and often depend on extensive hardware deployment. Advances in artificial intelligence and computer vision offer an opportunity to replace sensor-centric systems with lightweight, vision-based models capable of continuous surveillance. By analyzing real-time visual cues associated with electrical faults, these systems can provide early detection, precise localization, and contextual information, enabling faster decision-making and proactive maintenance in modern power distribution infrastructures.

II. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

Manual fault detection in power distribution networks is slow and often inefficient. It can lead to long power cuts, delays in repair work, and safety risks for maintenance workers. Linemen usually have to walk or travel along power lines and inspect each pole to find short circuits. This process takes a lot of time and can be dangerous, especially in high-voltage areas or bad weather conditions. When faults are not found quickly, it affects the reliability of the power supply and causes inconvenience to customers. Because of this, there is a clear need for an automated system that can detect and locate short circuits in real time. Such a system can send quick alerts, reduce power outage time, improve worker safety, and help maintenance teams fix problems faster and more efficiently.

Objectives

- **Real-Time Automated Short Circuit Detection:** Develop an AI-based system using the YOLOv11 model to continuously monitor live video streams and instantly detect visual indicators of short circuits such as sparks, smoke, and flashes.
- **Automated Alert System with Exact Fault Location:** Generate immediate fault notifications for the responsible lineman by providing the detected fault image, GPS-based location, and timestamp for faster and safer maintenance.
- **Visual Incident Reporting:** Capture and transmit real-time images of the detected short-circuit event along with location details through a web-based dashboard, enabling efficient fault verification

III. LITERATURE REVIEW

A. Traditional Manual Fault Detection Methods

Conventional short-circuit detection in power distribution networks primarily relies on manual inspection by linemen. This approach involves physical patrolling of distribution lines to locate faults, which is time-consuming, labor-intensive, and exposes workers to hazardous electrical and environmental conditions. Manual methods also result in delayed fault localization, contributing to prolonged power outages.

B. Sensor-Based and IoT Monitoring Systems

Recent studies have explored IoT-based fault detection using electrical sensors to monitor parameters such as current, voltage, and temperature. While these systems improve detection accuracy, they require extensive sensor deployment, stable communication networks, and frequent maintenance. High installation costs and limited scalability restrict their adoption, especially in rural or resource-constrained areas.

C. Lack of Real-Time Visual Fault Confirmation

Most existing fault detection systems provide numerical or alert-based notifications without visual evidence of the fault. The absence of real-time images limits situational awareness and may lead to misinterpretation of fault severity, increasing response time and inspection efforts.

D. Automated Alert and Localization Limitations

Several monitoring solutions fail to provide precise fault location details, often requiring manual verification even after fault detection. Inefficient alert mechanisms delay lineman response and increase outage restoration time, highlighting the need for accurate localization and structured alert delivery.

E. Need for Integrated AI-Based Detection Frameworks

Although recent research has improved fault detection using IoT devices and machine learning techniques, there is still a need for an integrated vision-based system that combines real-time AI detection, visual fault confirmation, location tracking, and automated alert generation. Such a framework can significantly reduce response time, improve lineman safety, and provide a cost-effective solution for modern power distribution monitoring.

IV. MATHEMATICAL MODELING

A. Notation

Let

$$F = \{f_1, f_2, \dots, f_T\}$$

be the sequence of video frames captured from the webcam over time $t = 1, 2, \dots, T$.

Let

$$C = \{c_1, c_2, \dots, c_M\}$$

represent the set of object classes, where the primary class is **Short-Circuit**.

Let

$$D_t$$

denote the set of detections at frame t . Each detection is represented as

$$d = (b, l, s)$$

where

- b = Bounding box
- l = Class label
- $s \in [0,1]$ = Confidence score

B. YOLOv11 Detection Model

The detection model is represented as

$$D_t = D(f_t; \theta)$$

where

- f_t is the input image frame,
- θ represents the trained YOLOv11 model parameters,
- D_t is the set of detected objects.

Each detection contains

$$D_t = \{(b_{t,j}, l_{t,j}, s_{t,j})\}_{j=1}^{n_t}$$

where

- $b_{t,j}$ = Bounding box
- $l_{t,j}$ = Predicted class
- $s_{t,j}$ = Detection confidence
- n_t = Number of detected objects.

C. Confidence Threshold and Event Detection

To eliminate low-confidence detections, a confidence threshold τ is applied.

$$\widetilde{D}_t = \{d \in D_t \mid s_{t,j} \geq \tau\}$$

Only detections satisfying

$$s_{t,j} \geq \tau$$

are considered valid short-circuit events.

If the detected class is

$$l = \text{Short-Circuit}$$

the system generates a fault event and proceeds with alert generation.

D. GPS-Based Location Acquisition

After detecting a short-circuit event, the system retrieves the geographical location using the connected mobile device.

The fault location is represented as

$$G = (\phi, \lambda)$$

where

- ϕ = Latitude

- λ = Longitude

These GPS coordinates are converted into a Google Maps link and attached to the generated alert for easy navigation by maintenance personnel.

E. Alert Generation

Once a valid short-circuit event is detected, the system generates an alert containing the fault details.

The alert record is represented as

$$A_t = \{\text{Type, Confidence, Location, Image, Timestamp}\}$$

where

- Type = Short Circuit
- Confidence = Detection confidence score
- Location = GPS coordinates
- Image = Captured fault image
- Timestamp = Detection time

The alert is immediately transmitted to the Flask-based dashboard for monitoring.

F. Event Logging

Every detected short-circuit event is stored in the event log for future analysis and maintenance.

The event log is updated as

$$L \leftarrow L \cup A_t$$

where

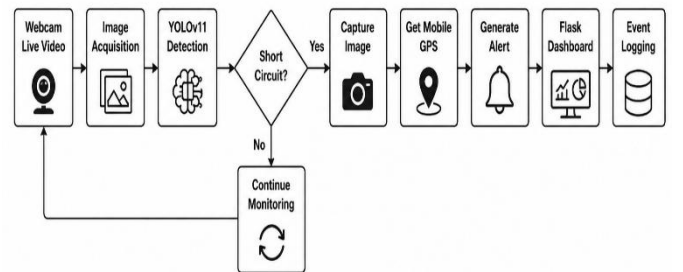
- L represents the historical event database.
- A_t is the newly detected alert.

The stored records include the fault image, location, timestamp, and detection details, enabling historical analysis and maintenance planning.

I. V. PROPOSED SYSTEM ARCHITECTURE

A. Overall Workflow

(Refer to Fig. 1)



A. Data Acquisition

The proposed system captures live video frames using a webcam connected to the monitoring system. These frames contain visual information such as sparks, smoke, and flashes produced during short-circuit events. Each frame is preprocessed using OpenCV, including resizing and normalization, before being passed to the YOLOv11 model for real-time detection. When a fault is detected, the system captures the detected image and obtains the current GPS coordinates from a connected mobile device.

B. Fault Detection Stream

A YOLOv11-based deep learning model is employed to analyze incoming video frames in real time. The model is trained on a custom dataset consisting of short-circuit images and normal electrical line conditions. It detects visual indicators such as sparks, flashes, and smoke by extracting spatial features and assigning confidence scores to each detected object. The detected fault region is highlighted using a bounding box, enabling accurate real-time identification of short-circuit events.

C. Fault Confirmation and Localization Stream

Detected fault events are validated using a predefined confidence threshold to minimize false detections. Once a short circuit is confirmed, the system captures the detected image and retrieves the current GPS coordinates from the connected mobile device. The captured fault image, location details, and timestamp are combined to generate a complete fault report, enabling quick identification of the affected location.

D. Alert Generation and Reporting Stream

When a fault is confirmed, the system automatically captures an image of the incident and generates a structured alert containing fault type, timestamp, and exact location details. These alerts are transmitted to a centralized web platform accessible to linemen and supervisory staff. Visual reporting improves situational awareness and assists linemen in preparing appropriate corrective actions before reaching the site.

E. Fusion and Decision Layer

The fusion layer combines detection confidence, fault localization data, time information, and historical event records to prioritize alerts. This layer ensures only critical and verified faults trigger notifications, avoiding alert fatigue. Priority levels guide maintenance response and enable efficient fault management across the distribution network.

VI. IMPLEMENTATION DETAILS

A. Software and Hardware

The proposed system is implemented using **Python** as the primary programming language. **OpenCV** is used for image acquisition and preprocessing, while the **YOLOv11** model is deployed using the **Ultralytics** framework built on **PyTorch** for real-time object detection. **Flask** serves as the backend framework for handling alerts and communication between the detection module and the dashboard. The frontend dashboard is developed using **HTML**, **CSS**, and **JavaScript** for real-time visualization of fault events. The system is executed on a standard computer with **8–16 GB RAM**, and GPU support can be utilized for faster model training and inference.

B. Training Pipelining

- 1) Collect real and simulated images representing short-circuit and normal electrical line conditions.
- 2) Annotate the dataset using Roboflow and Label Studio in YOLO format.
- 3) Preprocess the dataset through resizing and normalization.
- 4) Train the YOLOv11 model using the annotated dataset.
- 5) Validate the trained model using validation images and an appropriate confidence threshold.
- 6) Integrate the trained model with OpenCV, Flask, GPS, and the web dashboard for real-time fault detection and alert generation

C. Optimization

The YOLOv11 model is trained using the Stochastic Gradient Descent (SGD) optimizer with momentum. The model is trained for 100 epochs using an image size of 640×640 pixels. During inference, a confidence threshold of 8.5 is used to ensure reliable short-circuit detection while minimizing false positives. These parameters provide stable convergence and efficient real-time detection performance.

D. Loss Function

The YOLOv11 model optimizes a composite loss function consisting of bounding box localization loss, classification loss, and objectness (confidence) loss. These loss components enable accurate localization of short-circuit regions, reliable classification of fault events, and robust detection performance under varying environmental and lighting conditions.

VII. DATASETS

The proposed system is trained using a custom dataset consisting of images representing short-circuit events and normal electrical line conditions. Since no suitable public dataset was available for this application, the dataset was created by collecting real and simulated short-circuit images and video frames. All images were manually annotated using Roboflow and Label Studio in YOLO format. The dataset contains 200 images, which were divided into 160 training images (80%) and 40 validation/testing images (20%). Data augmentation techniques such as resizing, flipping, and brightness adjustment were applied to improve model generalization under different environmental and lighting conditions.

VIII. ADVANCED THEORETICAL FOUNDATIONS

A. Deep Learning-Based Object Detection: The proposed system is based on the principles of deep learning and computer vision for real-time object detection. The YOLOv11 model employs Convolutional Neural Networks (CNNs) to automatically extract spatial features from each input frame and detect short-circuit events in a single forward pass. This enables high-speed detection while maintaining reliable localization and classification performance.

B. Confidence-Based Fault Verification: Each detected object is assigned a confidence score by the YOLOv11 model. A predefined confidence threshold is applied to eliminate low-confidence detections and reduce false positives. When the confidence exceeds the threshold, the detected frame is considered a valid short-circuit event, and the system proceeds to generate an alert.

C. Real-Time Alert Generation: After fault verification, the system captures the detected image, retrieves the current GPS coordinates from the connected mobile device, and generates an alert containing the fault image, location, and timestamp.

IX. DETECTION AND ALERT WORKFLOW

The proposed system follows the workflow below:

- Live video frames are captured using a webcam.
- OpenCV preprocesses each frame before passing it to the YOLOv11 model.
- The YOLOv11 model analyzes each frame and detects short-circuit events in real time.
- When a fault is detected, the system captures the fault image and retrieves the GPS coordinates from the connected mobile device.
- Flask generates an alert containing the fault image, GPS location, and timestamp.

- The alert is displayed on the web dashboard, where linemen can monitor and manage reported fault events.

EXPERIMENTAL SETUP AND RESULTS

A. Experimental Setup

The proposed AI-Based Real-Time Short Circuit Detection System was implemented using Python, OpenCV, Ultralytics YOLOv11, PyTorch, and Flask. A custom dataset consisting of real and simulated short-circuit images was prepared and manually annotated using Roboflow and Label Studio in YOLO format. The dataset was divided into training and validation sets, and the YOLOv11 model was trained for 100 epochs using an input image size of 640×640 pixels. Data augmentation techniques, including image flipping, scaling, rotation, and brightness adjustment, were applied to improve the model's robustness under varying environmental and lighting conditions.

The trained model was integrated with a webcam for real-time monitoring. OpenCV continuously captures video frames, which are processed by the YOLOv11 model to detect short-circuit events. When a fault is detected, the system captures the fault image, retrieves the current GPS coordinates from the connected mobile device, and transmits an alert containing the fault image, location, and timestamp to a Flask-based web dashboard. The complete system was validated through real-time demonstrations using live webcam streams and simulated short-circuit scenarios.

B. Results

The proposed system successfully detected short-circuit events from real-time webcam video streams, recorded the detected fault image, obtained the GPS location through a connected mobile device, and generated alerts on the web dashboard. Each alert contains the detected fault image, pole identification, Google Maps location link, timestamp, and alert management options such as Checked, Remind, and Delete. The system demonstrated reliable performance during both live demonstrations and testing with short-circuit images and videos.

Detection Performance

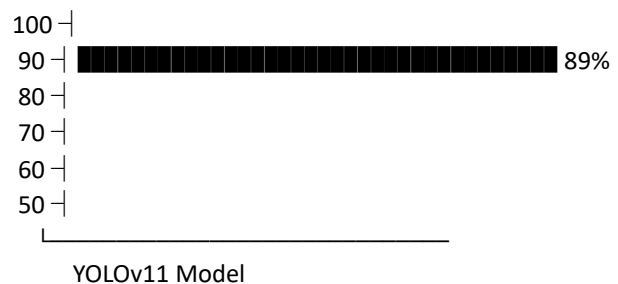


Fig. 3 shows the real-time detection of a short-circuit event using the trained YOLOv11 model. The model accurately identifies the fault region and displays the corresponding confidence score.

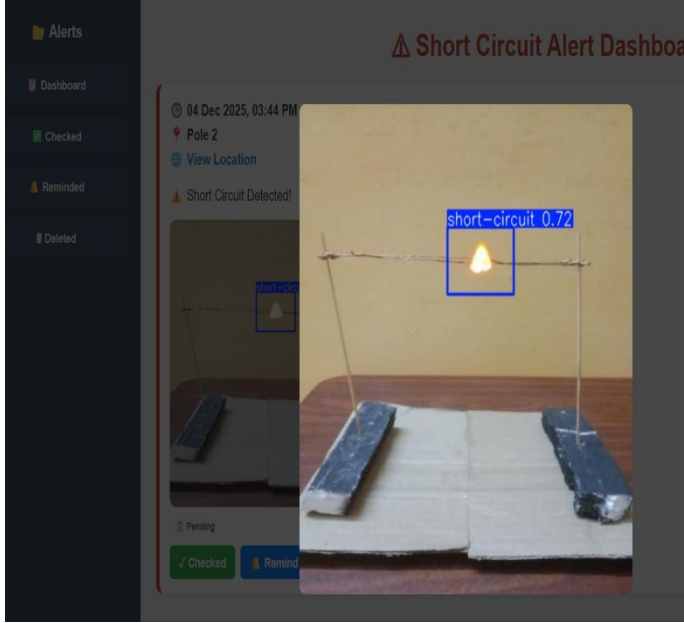


Fig. 3. Real-time short-circuit detection using the YOLOv11 model.

Figure 6 presents the Flask-based web dashboard developed for the proposed system. Whenever a short circuit is detected, the dashboard automatically displays the captured fault image, pole identification, Google Maps location, detection time, and alert management options.

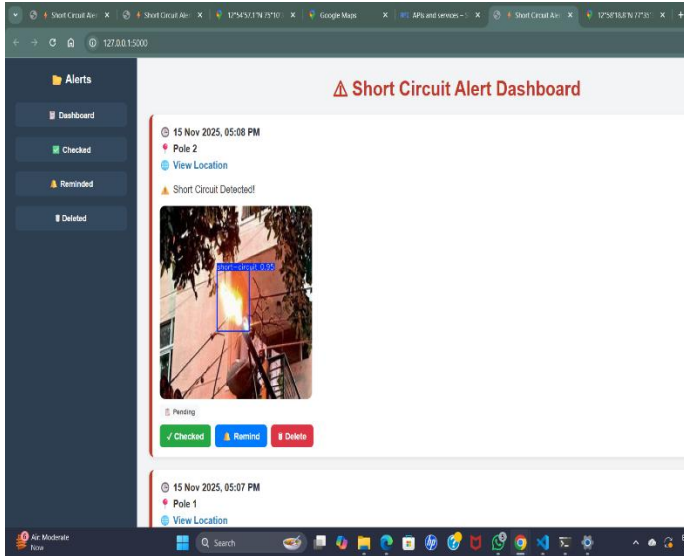


Fig. 4. Web-based dashboard displaying a detected short-circuit alert.

TABLE I. Experimental Configuration and Performance

Parameter	Value
Model	YOLOv11
Dataset	Custom Short-Circuit Dataset
Training Epochs	100
Image Size	640 × 640
Confidence Threshold	0.6
Framework	PyTorch (Ultralytics)
Backend	Flask
Image Processing	OpenCV
Detection Accuracy	89%

XII APPLICATIONS

The proposed system can be deployed across multiple power infrastructure environments:

- **Electric power distribution utilities:** Automated short-circuit detection and maintenance alert systems.
- **Smart grid infrastructures:** Integration with intelligent monitoring and predictive maintenance frameworks.
- **Industrial power networks:** Enhancing safety and reducing downtime in large facilities.
- **Rural electrification projects:** Faster fault resolution in remote or hard-to-access areas.

XIII .FUTURE WORK

Future work includes expanding the dataset with additional fault scenarios, improving low-light detection using infrared cameras, integrating cloud-based monitoring, developing a dedicated mobile application for linemen, implementing predictive fault analysis, and enhancing the dashboard with advanced analytics and historical fault visualization. Integration with smart grid platforms can further improve large-scale deployment and real-time monitoring.

XIII. CONCLUSION

The proposed AI-Based Real-Time Short Circuit Detection and Lineman Alert System addresses the limitations of conventional manual fault detection in electrical power distribution networks. In traditional methods, linemen are required to physically inspect multiple electrical poles to identify the fault location, resulting in increased response time, higher operational costs, and potential safety risks. To overcome these challenges, the proposed system employs a webcam for continuous real-time monitoring and utilizes the YOLOv11 deep learning model to automatically detect visual indicators of short-circuit events, such as sparks and flashes.

Whenever a short circuit is detected, the system captures the fault image, retrieves the current GPS location using a connected mobile device, and immediately sends an alert containing the fault image, location, timestamp, and fault information to a Flask-based web dashboard. The dashboard enables maintenance personnel to efficiently manage alerts through features such as Check, Remind, and Delete, thereby improving fault monitoring and maintenance operations.

The proposed system was successfully tested using a custom dataset and demonstrated reliable real-time detection performance during live demonstrations using images and videos. Compared with conventional inspection methods, the proposed solution reduces manual effort, improves response time, enhances lineman safety, and provides visual evidence for better decision-making. Overall, this work demonstrates the potential of integrating Artificial Intelligence, Computer Vision, and web technologies to develop an efficient, scalable, and cost-effective solution for intelligent fault monitoring in modern power distribution systems.

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